Rohan Malhotra

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EXPERIENCE

IBM Dallas, TX

Applications Developer Intern, IBM Consulting *May 2026 – Aug 2026*

- Fine-tuned Qwen-Coder-32B on Oracle EPM assets, raising EPM build-generation accuracy from **36.7% to 95.0%**.
- Built an on-prem EPM assistant that turns plain-language requests into reviewable JSON using RAG and MCP tools.
- Built an XGBoost cash-flow forecasting model and 13-day rolling cash forecasts in Oracle EPM Planning.

Robotics Engineer Intern *Jun 2026 – Aug 2026*

- Trained YOLO11 models on 898 hand-labeled images to **99.5% mAP@50** with 99.4% precision and 99.4% recall.
- Connected real-time inference to the Boston Dynamics Spot SDK via OpenCV, gRPC, and lock-free inference queues.
- Built an autonomous object-retrieval demo, presented it live at IBM DevCon, and open-sourced the full pipeline.

Aress Software New York, NY

Machine Learning Intern *Jun 2025 – Aug 2025*

- Trained and deployed PyTorch models on **500K+ multimodal records** across Azure and AWS.
- Designed NoSQL pipelines unifying 10+ data sources into one analytics layer; optimized SQL queries for speed.
- Built Power BI dashboards surfacing 20+ real-time KPIs for global operations teams; managed the JIRA workflow.

Kalshi New York, NY

Quantitative Developer *Jan 2026 – May 2026*

- Built Poisson hazard and Monte Carlo models of job-loss risk plus a Python/Next.js hedge-sizing engine.
- Wrote C++ risk tools, containerized the runtime with Docker, and deployed it to Render.

Hume Center for National Security and Technology, Virginia Tech Blacksburg, VA

Aerospace Researcher *Nov 2024 – May 2025*

- Programmed imaging and signal-processing tests in C for ContentCube, a CubeSat deployed from the ISS in 2025.

PROJECTS

EPM Wizard | *FastAPI, React, Groovy, Playwright* 2026

- Built a tool that turns plain-English requests into Oracle EPM forms and rules, with preview and approval steps.

Casen | getcasen.com | *JavaScript, OpenAI, Supabase, Voice AI* 2026

- Built a case-interview prep platform with frameworks, drills, M&A news, and voice-based AI mock interviews.

AI Quiz Maker | *Next.js, FastAPI, OpenAI* 2026

- Built an app that turns PDFs and notes into quizzes, flashcards, and recall prompts using an LLM.

GreenSticker | greensticker.us | *Python, OpenCV*

- Built a hands-free webcam mouse that tracks a colored sticker, with low-latency on-device processing.

RESEARCH

Reddit Data in Quantitative Financial Models (published) Jan 2025 – May 2025

Co-author, Virginia Tech (VTechWorks) *hdl.handle.net/10919/124730*

- Applied NLP sentiment scoring and regression models in Python to quantify sentiment's impact on equity returns.

EDUCATION

New York University, Courant Institute of Mathematical Sciences New York, NY

B.A. in Mathematics and Computer Science (Double Major) *Expected May 2027*

- **GPA:** 3.7/4.0 | Accelerated three-year degree | **Graduate course:** Mathematical Techniques for CS Applications
- **Coursework:** Machine Learning, Linear Algebra, Probability & Statistics, Numerical Methods, Algorithms

SKILLS

ML & AI: PyTorch, scikit-learn, XGBoost, YOLO11, OpenCV, LLM fine-tuning, LoRA, RAG, MCP, NLP, OpenAI API

Languages: Python, C++, C, SQL, Java, TypeScript, R, MATLAB

Data & Infra: NumPy, Pandas, SciPy, Linux, Docker, AWS, Azure, PostgreSQL, gRPC, FastAPI, Next.js, Power BI